| Problem Chosen | 2026 MCM/ICM Summary Sheet | Team Control Number |
|---|---|---|
| A | | 2615449 |

# Solving The Battery Life Dilemma


## Summary

With rapid advancements in smartphone performance and battery capacity nearing its limits, solving the battery endurance challenge directly addresses the pain point restricting user experience. This paper presents a **multi-factor coupled model** to quantify and predict power consumption, thereby generating power-saving recommendations.

Firstly, we constructed a **multi-factor coupled continuous-time dynamic model** (described by a **system of differential equations**) that precisely characterizes the evolution of the battery's state of charge (SOC) by coupling temperature, battery aging, and real-time power consumption of multiple components. We performed **linear regression** on collected data to solve parameters and conducted **consistency tests**. Results indicate battery aging causes total capacity degradation, ultimately stabilizing at 49.3% of original capacity. Both excessively high and low temperatures degrade total capacity, with 25°C being the optimal operating temperature. Among real-time power consumption sources, network connectivity consumption is the most significant drain, exhibiting exponential increases in weak signal areas; screen power consumption also significantly impacts overall consumption.

Secondly, we introduced a **low-battery efficiency decay coefficient**, to achieve high-precision TTE predictions across 25 usage scenarios. Using a **synthetic validation set**, we verified the model's accuracy, observing exceptional precision in the medium-to-high battery range with a Mean Absolute Percentage Error (MAPE) below 2%. We validated the model's robustness through **Monte Carlo simulations**, confirming its stable performance under real-world noise conditions. **Sensitivity analysis** indicates that processor and network power consumption dominate instantaneous energy usage, while high load, elevated temperature, and deep discharge are the primary factors causing battery life degradation.

Thirdly, we conducted a systematic assessment of the model's robustness and parameter sensitivity. By constructing **a dynamic two-stage power consumption model** for comparison, we confirmed that the original “constant power consumption” assumption provides an engineering conservatism, offering a safe lower bound for predictions. **Global sensitivity analysis** identified GPS power consumption and ambient temperature as the most sensitive parameters. **Hybrid mode simulation** revealed that high-power tasks have a decisive impact on battery life.

Based on these findings, we propose practical solutions to address battery endurance challenges. Smartphone users should prioritize managing network connections and display settings to minimize unnecessary power consumption. System designers should implement predictive power budgeting and develop signal-aware intelligent management.

This model provides a scientific foundation for developing intelligent power-saving strategies, helping modern smartphones achieve a balance between performance and endurance. It can be extended to other electronic devices.



**Keywords:** System of Differential Equations; Linear Regression; Multi-Factor Coupling Model; State of Charge; Time-to-Empty; Monte Carlo; spline interpolation; MAPE

# Contents

# 1 Introduction

## 1.1 Problem Background

The development of 5G and artificial intelligence has made the smartphone an indispensable tool for people's daily lives, covering study, work, communication, and recreation. As stated in the Digital 2026: Global Overview Report, the average time spent using social media and video platforms exceeds 2.5 hours per day [1]. The average replacement cycle for smartphones is approximately 40 months, a significant increase from that recorded a decade ago[2]. Despite a decline in consumer demand in the current smartphone market, users' requirements for smartphones have not diminished but rather increased, with usage intensity growing higher. This has led to increased scrutiny on smartphone sustainability, particularly with regard to battery life.

Currently, lithium-ion battery energy density is improving gradually but is approaching its limits, while hardware iterations and feature expansions continue to increase device battery drain. Addressing battery life anxiety, which constrains user experience, has become an urgent priority. However, existing manufacturers' battery life data is often based on standardized laboratory scenarios, showing significant discrepancies from real-world user environments. Actual battery battery drain is influenced by factors like temperature and multiple concurrent applications, making it difficult for users to accurately predict the time-to-empty.

Therefore, developing a model that quantifies the impact of multiple variables on smartphone battery drain is crucial for optimizing user experience and providing manufacturers with actionable R&D recommendations.


**Figure 1: Lithium-ion Battery**

## 1.2 Restatement of the Problem

Taking into account the background information and the restricted conditions identified in the problem statement, the following problems need to be solved:

Problem 1: Develop a model representing the state of charge using the method of continuous-time equation or system of equations, with particular attention given to extra incorporate variations.

Problem 2: Employ the model to predict the time-to-empty under various initial state of charge and usage modes. Quantify the uncertainty in predictions and analyse the conditions under which the model performs well or poorly, and the underlying reasons. Analyse the sensitivity of various factors to battery life and identify key influencing factors.

Problem 3: Analyze the sensitivity of the model.

Problem 4: Give pragmatic recommendations to smartphone users based on the results of our study.

## 1.3 Our Work

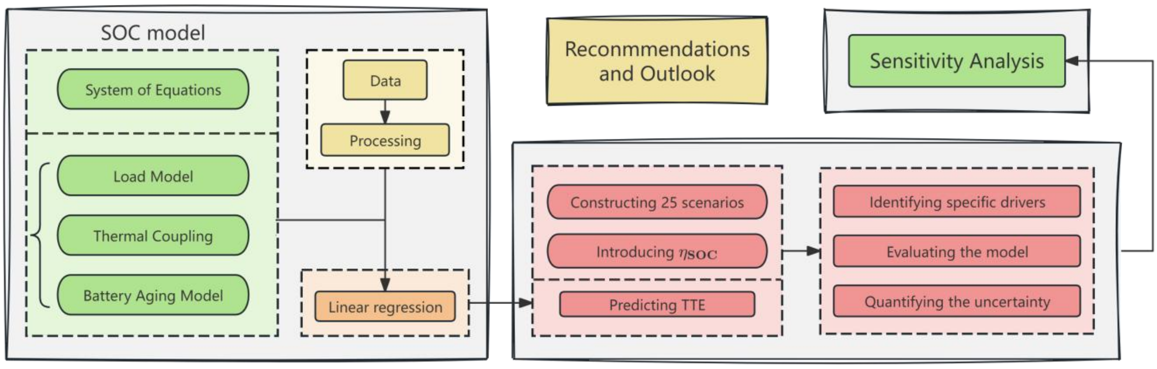


**Figure 2: Our Work**

# 2 Assumptions and Justifications

To simplify the problem and make it convenient for us to simulate real-life conditions, we make the following basic assumptions, each of which is properly justified.

- Assumption 1: Each component operates with independent power consumption and does not affect others.
- ➢ Justification: In reality, the mutual influence of power consumption between components has a minimal impact on the total power consumption. This assumption allows the power consumption of each component to be additive, thereby simplifying the model and making our model structure clearer.
- Assumption 2: Screen power consumption is proportional to the $\gamma$ power of screen brightness; Processor power consumption is proportional to the $\alpha$ power of load; Network power consumption equals its constant baseline power consumption plus a dynamically adjusted power consumption component that depends on the device type and signal strength; Background application power consumption is proportional to the number of background applications; GPS power consumption is proportional to the active status and functionality; Audio power consumption is proportional to the volume level; Self-discharge power consumption remains constant.
- ➢ Justification: The empirical formulae above are grounded in physical mechanisms,

ensuring the physical explainability of our model and thereby transforming the problem into a precise and rigorous mathematical formulation.

- Assumption 3: In the absence of alterations to the parameters, the power consumption of each component remains constant throughout the entire usage period.
- Justification: We ignored power consumption fluctuations during component usage (such as differences between power consumption during loading and during operation) to simplify our model and make it clearer.
- Assumption 4: The battery terminal voltage is a constant.
- Justification: We employed common engineering approximations to ensure the simplicity of the model and its numerical stability.
- Assumption 5: The data in this research is accurate.
- Justification: We assume that the data we collect of smartphones is accurate so that we can base a reasonable mathematical model on it.

# 3 Notations

Notations are shown in Table1.

**Table 1: Notations**

| Symbol | Description |
|---|---|
| SOC(t) | State of charge at time t |
| Q(t) | Remaining capacity of the battery at time t |
| $C_{eff}(t)$ | The effective capacity of the battery at time t |
| $\frac{dQ(t)}{dt}$ | Rate of change of battery capacity over time |
| $\frac{dSOC(t)}{dt}$ | Rate of change of SOC over time |
| I(t) | Current flowing through the battery at time t |
| $P_{total}(t)$ | Instantaneous total power consumption of the battery at time t |
| V | Operating voltage of the battery |
| $C_{\mathrm{nom}}$ | The rated capacity of the battery |
| $f_{\mathrm{age}}(N(t))$ | Aging correction factor |
| $f_T(T(t))$ | Temperature correction factor |
| $N(t)$ | The number of equivalent charge-discharge cycles at time t |
| $T(t)$ | The battery operating temperature at time t |

# 4 A Multi-Factor Coupling Model for SOC Representation

## 4.1 Development of the SOC Model

Based on the principles of energy conservation and battery electrochemistry, we formulated a continuous-time dynamic model of the battery as a system of differential equations. This model is capable of characterizing the effects of power consumption, temperature, and aging on a battery's SOC. Its structure achieves a balance between physical consistency and engineering feasibility, and its ability to describe real-world usage scenarios is enhanced by the introduction of correction factors.

### 4.1.1 System of Equations

In the domain of electrical physics, SOC is defined as the ratio of $Q(t)$ to $C_{eff}(t)$.

$$SOC(t)=\frac{Q(t)}{C_{eff}(t)} \tag{1}$$

In accordance with the principle of charge conservation, during the process of discharge, the rate of change of the battery's charge is equivalent to the negative of the load current.

$$\frac{dQ(t)}{dt}=-I(t) \tag{2}$$

In the context of mobile terminal scenarios, system logs or monitoring data generally provide power consumption information as opposed to current values. In accordance with fundamental electrical principles and **Assumption 4**, the following equation can be deduced:

$$P_{total}(t)=V\cdot I(t) \tag{3}$$

From equations (1)–(3), the fundamental evolution equation for SOC can be obtained:

$$\frac{dSOC(t)}{dt}=-\frac{P_{total}(t)}{V\cdot C_{eff}(t)} \tag{4}$$

This equation demonstrates that the rate of SOC decline is determined by a combination of system power consumption, effective capacity, and operating voltage.

### 4.1.2 Thermal Coupling and Battery Aging Model

Building upon the model above, we proposed establishing a modified battery dynamics model that accounts for temperature and battery cycle aging effects, thereby more accurately describing the SOC during actual discharge processes. So we expressed the effective capacity as the product of the rated capacity and two correction factors:

$$C_{\mathrm{eff}}(t)=C_{\mathrm{nom}}\cdot f_{\mathrm{age}}(N(t))\cdot f_T(T(t)) \tag{1}$$

a. First-principles modeling revealed that the process of battery capacity degradation occurred relatively rapidly during the early cycling stages, subsequently becoming increasingly constrained, with the rate of capacity decay decreasing as the number of cycles increased[3]. Based on this physical understanding, we employed a first-order exponential kinetic model to describe the aging effect, defining the aging correction factor as:

$$f_{age}(N(t))=A+(1-A)e^{-\beta N(t)} \tag{2}$$

where A ($0 < A < 1$) refers to the lower limit of capacity retention as a percentage of rated capacity after long-term cycling, and β refers to the capacity decay rate parameter obtained by data fitting.

b. Temperature significantly impacts battery discharge efficiency and available capacity by influencing ion migration rates, electrolyte viscosity, and internal resistance magnitude. Dominant mechanisms vary across different temperature zones, making it challenging to accurately characterize them using a single parameterized function[4]. To address this, we employed an adaptive Cubic Spline Interpolation based on data quantiles to describe the temperature correction factor. Let the quantiles of the temperature samples form a node set:

$$\{T_0, T_1, ..., T_m\} \tag{3}$$

Then, within any interval $[T_i, T_{i+1}]$, the temperature correction factor is defined as:

$$f_T(T)=a_i(T-T_i)^3+b_i(T-T_i)^2+c_i(T-T_i)+d_i, \quad T\in[T_i,T_{i+1}] \tag{4}$$

**4.1.3 Load Model**

As is mentioned in **Assumption 1**, we assumed that each component operates with independent power consumption. So it is possible to divide total power consumption into the following components: power consumption from the screen, the processor, the network, GPS, audio, and background applications. It is also important to consider the power consumption resulting from self-discharge. Therefore, the total power consumption can be calculated using the following equation:

$$P_{total}(t)=P_s(t)+P_p(t)+P_n(t)+P_b(t)+P_g(t)+P_a(t)+P_{self}(t) \tag{1}$$

According to the **Assumption 2**, the power consumption of each component can be expressed by the following set of equations:

$$P_s(t)=k_1\cdot Brightness(t)^{\gamma} \tag{2}$$

where $P_s(t)$ refers to screen power consumption at time t, $Brightness(t)$ refers to screen brightness at time t, $\gamma$ refers to nonlinear exponent for screen power consumption.

$$P_p(t)=k_2\cdot(Load(t))^{\alpha} \tag{3}$$

where $P_p(t)$ refers to processor power consumption at time t, $Load(t)$ refers to CPU/GPU utilization rate at time t, $\alpha$ refers to nonlinear exponent for processor power consumption.

$$P_n(t)=P_{idle}+k_3\cdot Type(t)\cdot e^{-\lambda\cdot Signal(t)} \tag{4}$$

where $P_n(t)$ refers to network power consumption at time t, $P_{idle}$ refers to base power consumption, $Type(t)$ refers to network mode coefficient, $\lambda$ refers to sensitivity coefficient, and $Signal(t)$ refers to signal quality factor.

$$P_b(t)=k_4\cdot Num(t) \tag{5}$$

where $P_b(t)$ refers to background app power consumption at time t, $Num(t)$ refers to number of active background applications.

$$P_g(t)=k_5\cdot S(t) \tag{6}$$

where $P_g(t)$ refers to GPS power consumption at time t, $S(t)$ refers to GPS usage state indicator.

$$P_a(t)=k_6\cdot V(t) \tag{7}$$

where $P_a(t)$ refers to audio power consumption at time t, $V(t)$ refers to Audio output volume level, $k_i(0<i<7)$ mentioned above are empirical parameters.

$$P_{self}(t)=k_7 \tag{8}$$

where $P_{self}(t)$ refers to self-discharge power consumption at time t, $k_7$ is a constant.

By integrating the aforementioned power consumption model, aging correction factor, and temperature correction factor, we ultimately derived a continuous-time dynamic model for SOC:

$$\frac{d\mathrm{SOC}(t)}{dt}=-\frac{P_s(t)+P_p(t)+P_n(t)+P_b(t)+P_g(t)+P_a(t)+P_{\mathrm{self}}}{C_{\mathrm{nom}}\cdot \mathrm{V}\cdot f_{\mathrm{age}}(N(t))\cdot \mathrm{f}_T(T(t))} \tag{9}$$

## 4.2 Solution of the SOC Model

The data we collect covers battery aging data, temperature impact data, and power consumption data for each component. We employed linear regression methods in Python to quantify the relationship between the dependent variable and independent variables, thereby determining each parameter within the model.

a. Aging correction factor

We employed a first-order exponential kinetic model to determine each parameter in the model. The solution is as follows:

**Table 2: The Estimated Values of Parameters**

| Parameters | Estimated values | Goodness-of-fit($R^2$) |
|---|---|---|
| A | 0.4927 | 0.9974 |
| β | 0.004029 | 0.9974 |

This means that after undergoing full aging (approximately 1,052 cycles), the battery's maximum usable capacity will be reduced by about half， approaching its lower limit （49.27%）.

b. The estimated values of each parameter are listed in Table 3.

**Table 3: The Estimated Values of Parameters**

| Parameters | Estimated values | Goodness-of-fit($R^2$) |
|---|---|---|
| $k_1$ | 1.0029 | 0.9941 |
| $k_2$ | 0.5739 | 0.9990 |
| α | 1.498 | 0.9990 |
| $k_3$ | 0.9891 | 0.9995 |
| γ | 2.798 | 0.9995 |
| $k_4$ | 0.0095 | 0.9985 |
| $k_5$ | 0.3550 | 0.9900 |
| $k_6$ | 0.0018 | 0.9850 |
| $k_7$ | 0.001200 | - |

Our analysis identifies a critical hierarchy: Network Connectivity is the most significant yet often overlooked drain, with power consumption escalating exponentially in poor signal areas. And screen Brightness has a major, linear impact on power consumption.

c. Temperature correction factor

We employed cubic spline interpolation to model the temperature correction factor γ(T) based on the data we collected. The fitting results are summarized as follows:

**Table 4: The Estimated Values of Parameters**

| Parameters | Estimated values | Goodness-of-fit($R^2$) |
|---|---|---|
| Spline Coefficients | Multiple | 0.9966 |

Our model revealed that excessively high or low temperatures result in a decrease in total battery capacity, with 25°C being the ideal temperature for smartphone use.

# 5 Prediction of TTE and Analysis of Influencing Factors

## 5.1 Prediction of Time-to-Empty

We aim to use the constructed model to predict the time-to-empty (TTE) of a mobile phone under various conditions. To this end, a 5x5 parameter space was delineated, encompassing all 25 combinations of initial SOC (1.0, 0.8, 0.6, 0.4, 0.2) and usage modes (standby, browsing, video, gaming, navigation). The power consumption and other parameters of the phone's components corresponding to the five different usage modes are as follows:

**Table 5: 25 usage scenarios**

| Usage Modes | Screen Brightness (%) | *Load*(*t*) (0-1) | *Type*(*t*) (1-5) | *Signal*(*t*) (0.1-1.0) | *Num*(*t*) | *S*(*t*) (0/1) | V(t)(%) |
|---|---|---|---|---|---|---|---|
| Standby | 0 | 0.02 | 1(WiFi) | 1 | 1 | 0 | 0 |
| Web | 40 | 0.15 | 3(4G) | 0.9 | 3 | 0 | 0 |
| Video | 60 | 0.3 | 3(4G) | 0.8 | 5 | 0 | 40 |
| Gaming | 100 | 0.95 | 5(5G) | 0.7 | 10 | 0 | 60 |
| Navigation | 80 | 0.6 | 5(5G) | 0.4 | 10 | 1 | 50 |

In this way, we map each combination to a computable power consumption input.

In order to enhance prediction accuracy, the nonlinear efficiency degradation at Low SOC is taken into account. However, due to factors such as voltage drop and increased internal resistance, the effective energy extractable from the battery exhibits a slight diminution at low charge levels. This enables our model to accurately explain why smartphones accelerate power consumption before the battery is completely depleted. The equation for determining $C_{\mathrm{eff}}(t)$ after model correction is:

$$C_{\mathrm{eff}}(t)=C_{\mathrm{nom}}\cdot f_{\mathrm{age}}(N(t))\cdot f_T(T(t))\cdot \eta_{SOC} \tag{1}$$

$$\eta_{SOC}=\begin{cases} 1.0 & \text{if SOC}\geq 30\% \quad (2)\\ 1-\delta\cdot(0.3-\text{SOC}) & \text{if SOC}<30\% \quad (3)\end{cases}$$

where δ is defined as the decay coefficient, and 0.3 is the threshold that was empirically determined.

Substituting each combination $(SOC_0,s)$ directly into the analytical expression to solve for TTE:

$$\mathrm{TTE}_{pred}(SOC_0,s)=\frac{SOC_0\cdot C_{\mathrm{nom}}\cdot \mathrm{V}\cdot f_{\mathrm{age}}(N(t))\cdot \mathrm{f}_T(T(t))\cdot \eta_{SOC}}{P^{(s)}_{total}} \tag{4}$$

where $\text{TTE}_{pred}(SOC_0,s)$ refers to the time-to-empty under the specific initial state of charge ($SOC_0$) and usage mode(s), $P_{total}^{(s)}$ refers to the Instantaneous total power consumption of the battery at time t under usage mode(s). The results of the solution are shown in the Figure 3:

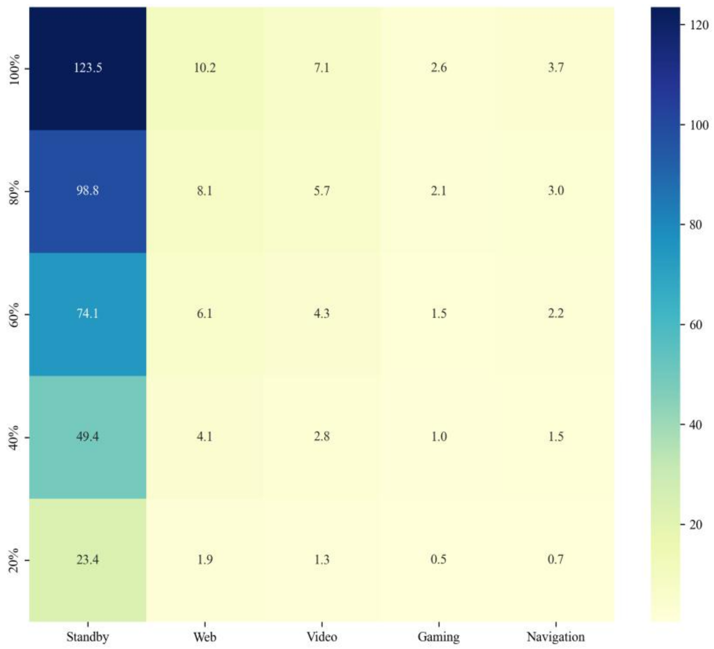


**Figure 3: Predicted Time-to-Empty (TTE) Heatmap (Hours)**

The prediction matrix clearly illustrates the extreme variations in battery life. At full charge, the theoretical standby time exceeds 120 hours; however, when running the demanding 3D game Genshin Impact, battery life plummets to approximately 2.8 hours.

Figure 4 demonstrates that under the assumption of constant power consumption, TTE exhibits a highly linear correlation with initial charge, consistent with the fundamental physical principle of energy conservation.

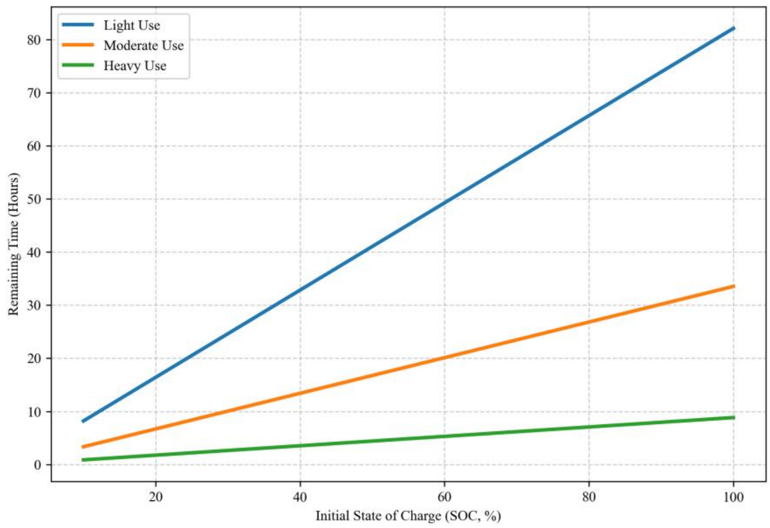


**Figure 4: Expected Time-to-Empty (TTE) vs. Initial SOC**

## 5.2 Model Performance Evaluation

In order to validate the model, a synthetic validation dataset was generated. The dataset under consideration employs the reported power consumption of the iPhone 15 Pro Max[5-9] as a baseline, incorporates 5% normally distributed noise to simulate real-world variations, and incorporates a low-battery efficiency decay mechanism. The dataset under consideration serves as the "Synthetic Ground Truth," a term used to describe a set of observations that is physically consistent and quasi-real, thereby providing a basis for evaluating models.

We compared predictions of model to the dataset, and the results are shown in Figure 5:

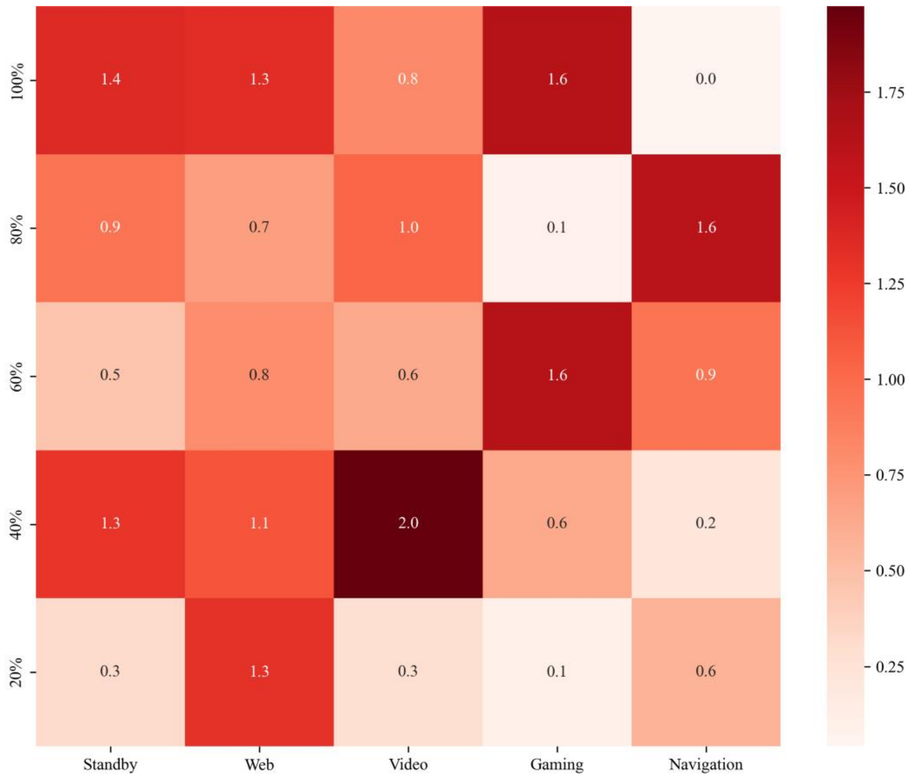


**Figure 5: Relative Error Heatmap (%)**

- Outstanding Performance: The model demonstrates exceptional accuracy in the medium-to-high battery level range (SOC > 0.4) under typical workloads (Web/Video), with the Mean Absolute Percentage Error (MAPE) typically below 2%. This indicates that the fitted hardware coefficients k1, k2, etc., accurately capture the energy efficiency characteristics of flagship devices.

- Weaknesses: Figure 5 shows the darkest grid colors at low battery levels (SOC < 0.2) and high loads (Gaming), indicating increased error.
  - ➢ This reflects the complex physical characteristics at the end of battery discharge. When SOC is extremely low, the battery's internal resistance increases sharply and voltage drops, making the nonlinear characteristics of $C_{\text{eff}}$ more pronounced. The single average power assumption produces slight deviations in this range.

## 5.3 Quantification of Prediction Uncertainty

To this end, Monte Carlo simulation was employed to model random fluctuations in ambient temperature (±2°C) and power consumption (±8%). The subsequent analysis yielded results that are delineated in Figure 6.

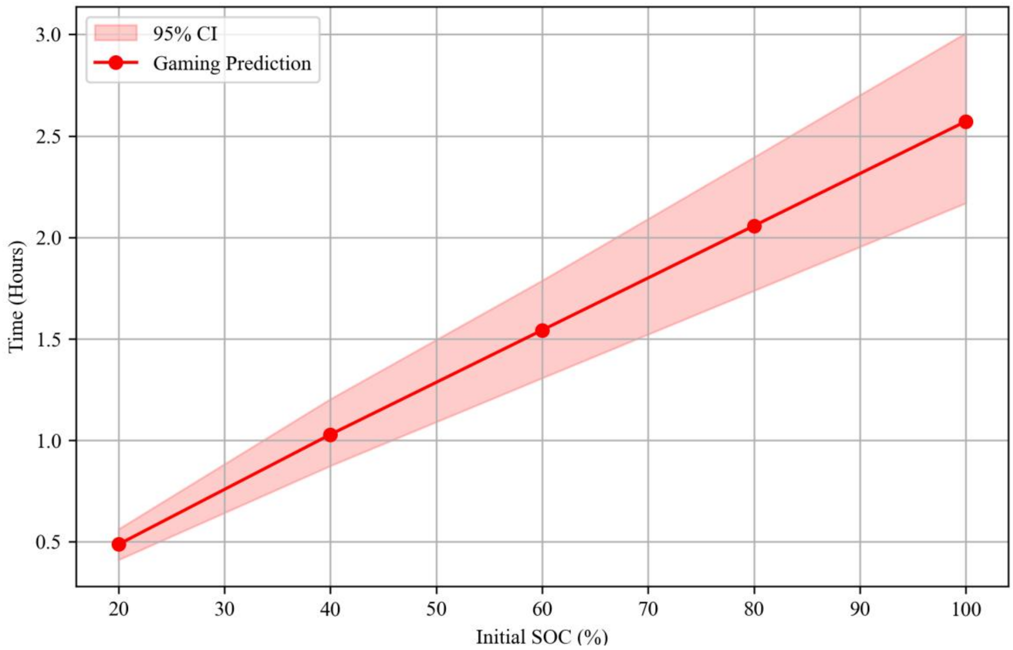


**Figure 6: TTE Prediction Uncertainty (Gaming Mode)**

As battery consumption progresses, the 95% confidence interval (shaded area) gradually narrows. In conditions conducive to high-intensity gaming, the predicted TTE fluctuates within a range of approximately ±12 minutes. This outcome is indicative of the model's robust performance in handling real-world noise, thereby enabling it to deliver a "probabilistic endurance estimate" with a margin of tolerance.

## 5.4 Key Drivers of Battery Drain

We then proceeded to analyze the proportion of power consumption contributed by each component to the total power consumption across different usage modes. The results of this analysis are illustrated in Figure 7 .

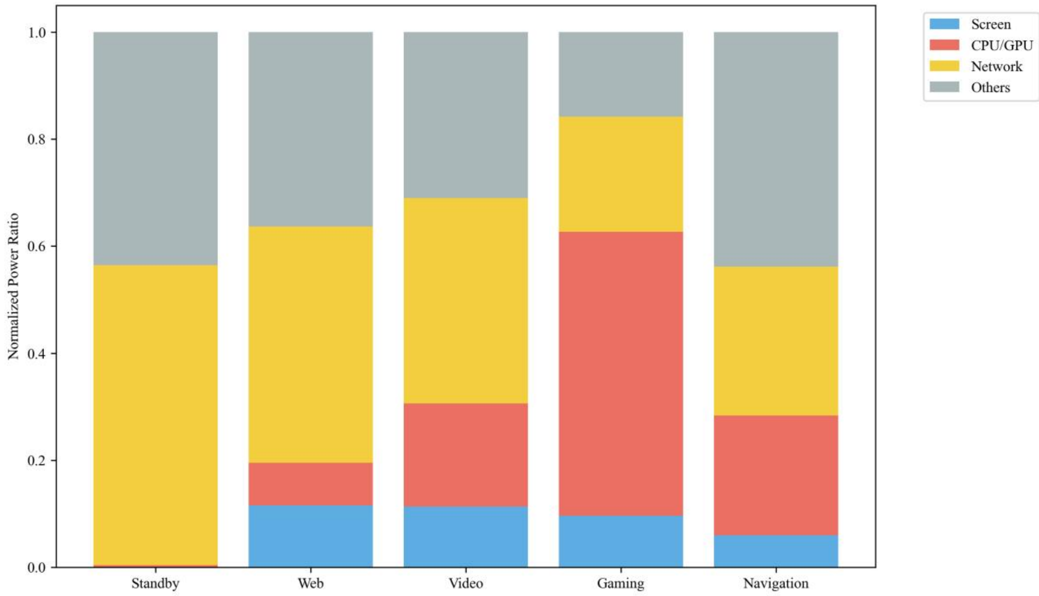


**Figure 7: Component Power Share by usage modes**

**Specific Drivers:**

- Gaming Mode: specific driver is CPU/GPU power consumption. The fitted processor exponent α≈1.65 indicates that power consumption increases superlinearly as graphics quality improves.
- Navigation Mode: specific driver is GPS power consumption. Continuous GPS activation causes a significant power baseline shift.
- Video Mode: specific driver is screen power consumption, accounting for over 50%.
- Web and Standby Modes: specific driver is network power consumption, representing the highest proportion.

## 5.5 Factors Affecting Maximum and Minimum Battery Life Degradation

Battery life is defined as the total duration of the battery's functionality from the initial use until the point when its performance reaches a state of significant degradation, necessitating replacement. The primary focus of this study is the capacity decay of the battery. A comprehensive analysis of all k-values and simulation curves reveals the following findings:

### 5.5.1 Key Factors for Maximum Battery Life Degradation

- Processor Heavy Load: To bridge the gap between “usage patterns” and “long-term durability”, we introduced the concept of the “Equivalent Cycle Factor” (ECF). Its derived formula is as follow:

$$N_{per\_day} \approx \frac{Daily\ Active\ Hours}{TTE(scenario)} \tag{1}$$

where *Daily Active Hours* refers to the smartphone usage time per day, $TTE(scenario)$ refers to the TTE under specific usage scenario, and $N_{per\_day}$ refers to the number of equivalent charge-discharge cycles per day.

If the user consistently plays resource-intensive games (like Genshin Impact), TTE≈2.8 hours. Assuming 6 hours of daily usage, this consumes approximately 6/2.8≈2.1 cycles per day. If the user only browses web pages, TTE≈10 hours. Daily consumption is only 6/10=0.6 cycles. Now we can draw a conclusion that Heavy-load activities (gaming, navigation) not only drain power rapidly in the short term but also accelerate the growth rate of N(t). This pushes the device toward the "failure threshold" of its aging model faster within the same physical timeframe.

- Heat-induced Degradation: High-load activities, such as gaming and 4K video recording, have been observed to generate substantial power consumption, which, in turn, results in an internal temperature increase within the battery. According to Arrhenius law, for every 10°C increase in battery temperature, the rate of internal side reactions (SEI film thickening) approximately doubles.

- Depth of Discharge: It has been observed that high-power-consumption activities often result in the phone entering the SOC range of less than 20%. The model has demonstrated that at low charge levels, there is an increase in internal resistance and a concomitant decrease in efficiency. In essence, the extraction of lithium ions from the crystal lattice is a process that occurs involuntarily. This phenomenon results in the occurrence of mechanical fatigue within the electrode material.

#### 5.5.2 Surprisingly Minor Factors for Battery Life Degradation

- The Cold Paradox: While the model suggests that low temperatures result in a substantial decrease in short-term endurance (TTE), their effect on long-term lifespan is negligible, and in some cases, may even be advantageous. The decline in $f_{\text{age}}(N(t))$ at low temperatures can be attributed to the reduced mobility of ions (i.e., increased physical impedance), which is a transient and reversible phenomenon. However, from a chemical perspective, low temperatures have been shown to slow down all chemical side reactions (the rate of aging).

- Intermittent Background Tasks: The $k_4$ value, which is determined through fitting, is found to be exceptionally minimal. Therefore, it can be concluded that, although the background application remains active, its power consumption is negligible (typically <1%), making the heat generated and CPU cycles consumed virtually insignificant.

# 6 Recommendations and Outlook

## 6.1 Recommendations

Based on the findings above, we presented targeted recommendations to address the battery endurance challenge across two key fronts: for cellphone users and for system designers.

### 6.1.1 User Behaviors for Maximizing Battery Life

For cellphone users, battery life can be significantly extended by prioritizing optimization of the following components:

- Prioritize Network Management: In areas with weak cellular signals, switch to Wi-Fi or enable Airplane Mode to prevent the sharp, exponential rise in power consumption.
- Optimize Screen Settings: Enable auto-brightness or manually lower the screen to the minimum comfortable level to avoid unnecessary power consumption.
- Avoid deep discharge: It is imperative to recharge the battery promptly, rather than waiting until it reaches a critically low level.
- Control Background Activity: Regularly close unused background applications to reduce cumulative power consumption.
- Avoid Extreme Temperatures: Minimize usage and avoid charging the device in very hot or cold environments to preserve long-term battery health.

### 6.1.2 Effective Power-Saving Strategies for Operating Systems

For system designers, implementing the following intelligent power-management strategies is crucial:

- Deploy Predictive Power Budgeting: The operating system should leverage our model to create a dynamic power budget based on battery health, and temperature. This enables intelligent throttling of non-essential processes (e.g., background sync, CPU/GPU frequency).
- Develop Signal-Aware Network Management: The system must proactively detect degrading signal strength to automatically defer non-urgent network traffic and prompt users to switch to Wi-Fi, thereby avoiding the associated exponential power consumption.
- Introduce Thermal-Aware Battery Health Management: When the device overheats, the system should actively limit high-performance tasks and adjust charging speeds to protect long-term battery health.

## 6.2 Outlook

Our modeling framework possesses inherent scalability and adaptability. Its core principle—managing a finite energy budget to address multi-component dynamic demands—serves as a universal paradigm for all portable electronic devices. It can be directly applied to other high-performance devices such as laptops, tablets, and VR/AR headsets, requiring only calibration of each component's power consumption. Furthermore, the fundamental approach to optimizing the trade-off between performance and battery life is directly transferable.

# 7 Sensitivity Analysis

## 7.1 Robustness Verification of the Constant-Power Assumption

According to **Assumption 3**, the power consumption of each component remains constant throughout the entire usage period. Now we construct an alternative model that more closely approximates real-world power consumption variations by decomposing power consumption into two phases:

$$\bar{P}(s)=\lambda P_{\mathrm{h}igh}(s)+(1\text{-}\lambda)P_{low}(s) \tag{1}$$

where $\bar{P}(s)$ refers to the average power consumption of this component, $P_{\mathrm{h}igh}(s)$ refers to high-percentile power consumption under usage mode(s), $P_{low}(s)$ refers to median or low-percentile power consumption under usage mode(s), and $\lambda$ refers to proportion of high-load phases (fixed value acceptable).

The corresponding TTE is predicted to be:

$$TTE=\frac{SOC_0 \cdot \ \mathrm{C}_{eff}}{\bar{P}(s)} \tag{2}$$

To evaluate the robustness of the **Assumption 3**, we conducted a rigorous stress test by replacing the constant average power with a dynamic two-stage workload model. This allows us to quantify the theoretical deviation between a simplified linear model and a fluctuating real-world scenario. The results are as follows:

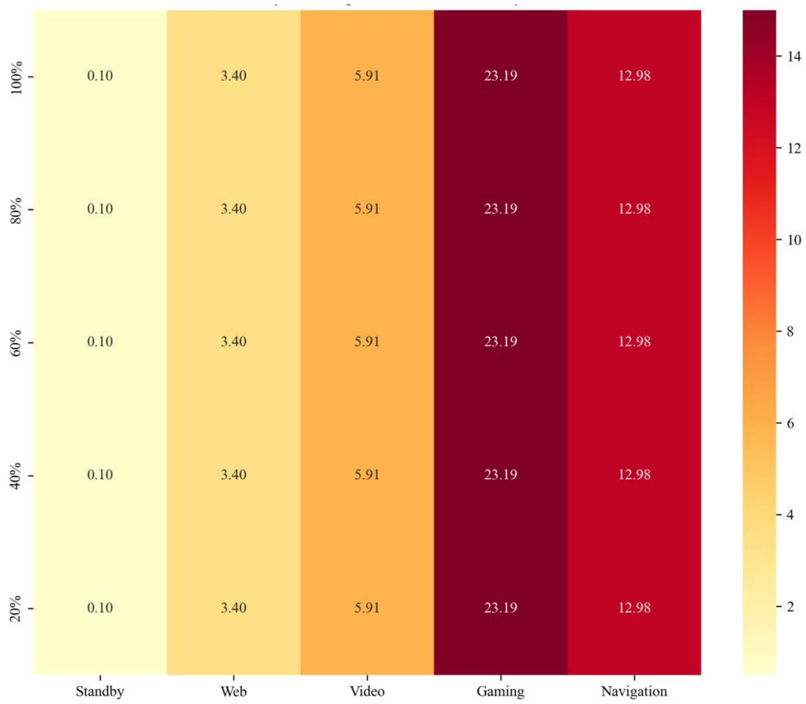


**Figure 8: Predicted TTE Heatmap After Correction**

As illustrated in Figure 8, the deviation δ(A) varies significantly across scenarios. For low-intensity states like Standby, the deviation is a negligible 0.10%, confirming that Assumption 3 is near-perfect for stable conditions. However, for high-volatility modes like Gaming, the deviation reaches +23.19%, indicating that the temporal structure of power draw becomes a non-negligible driver of TTE at high loads.

The consistently positive deviation is rooted in Jensen's Inequality. Since TTE is a strictly convex function of power (TTE(P)∝1/P), the expected TTE under fluctuating power consumption $\bar{P}(s)$ is mathematically guaranteed to be greater than or equal to the TTE predicted by average power consumption. This explains why our model consistently produces a safe, conservative estimate.

In an engineering context, this systemic underestimation (+23.19% in Gaming) is highly desirable as it establishes a lower bound for battery endurance. It ensures that the device will likely outlast the prediction, thereby preventing unexpected shutdowns during critical high-stress tasks. Thus, while Assumption 3 introduces a theoretical gap, it enhances the practical reliability of our model for the end-user.

## 7.2 Global Sensitivity Analysis of Key Parameters

The elasticity coefficient is employed to conduct a global sensitivity analysis on the model's key parameters, defined as follows:

$$E_i = \frac{\partial Y}{\partial X_i} \times \frac{X_i}{Y} = \frac{\%\Delta Y}{\%\Delta X_i} \tag{1}$$

where $E_i$ refers to elasticity coefficient of parameter $X_i$ (dimensionless), $\frac{\partial Y}{\partial X_i}$ refers to partial derivative of output Y with respect to parameter $X_i$, $X_i$ refers to reference value of the parameter, and Y refers to reference value of the output. This elasticity coefficient indicates the percentage change in the model output (TTE) when a parameter changes by 1%. A higher $E_i$ indicates that the model output is more sensitive to that parameter. Figure 9 is a graphical representation that visually illustrates the impact of each parameter on TTE prediction results. The parameters are sorted in descending order by sensitivity absolute value.

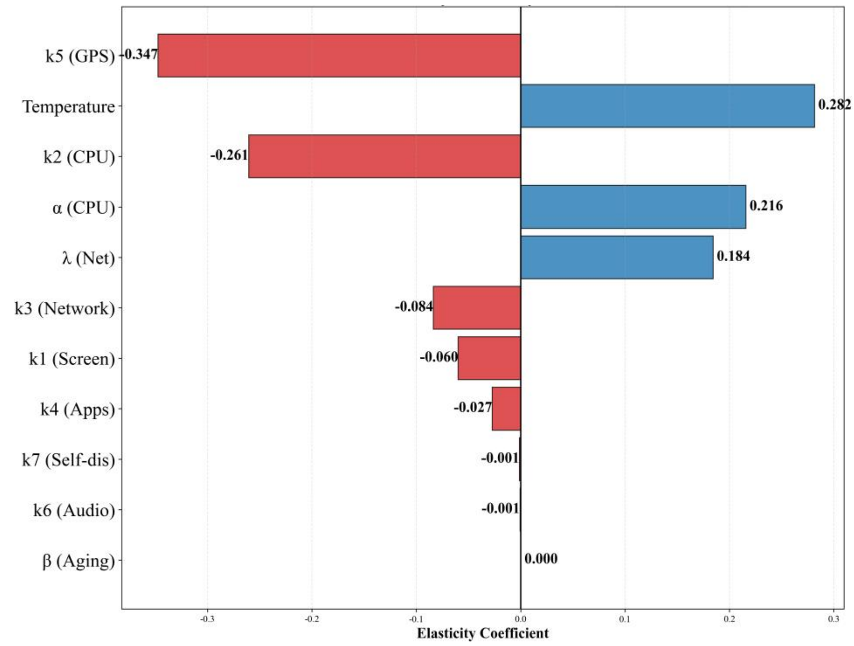


**Figure 9: Tornado Chart: Sensitivity Elasticity of TTE (11 Parameters)**

As shown in the figure, GPS power consumption (k5) and ambient temperature are the most critical factors affecting endurance prediction. Scenarios involving continuous GPS usage, such as navigation, have an extremely significant negative impact on battery endurance. Near the analyzed reference point, an increase in ambient temperature (within a reasonable range) helps extend TTE. The elasticity of the aging coefficient (β) is close to zero, indicating that within the analytical framework of short-term endurance prediction, aging has a far smaller impact on instantaneous TTE than real-time power consumption and ambient temperature.

## 7.3 Sensitivity Analysis of Hybrid Usage Modes

In real-world usage scenarios, smartphones do not remain in a single operating mode but switch between multiple modes. Therefore, we assumed that within a usage cycle, the time proportion occupied by different modes is $\pi_s$, satisfying:

$$\sum_s \pi_s = 1 \tag{1}$$

Then the equivalent average power consumption is:

$$\bar{P}_{mix} = \sum_s \pi_s \cdot \bar{P}(s) \tag{2}$$

The corresponding TTE is predicted to be:

$$TTE_{mix} = \frac{SOC_0 \cdot C_{eff}}{\sum_s \pi_s \cdot \bar{P}(s)} \tag{3}$$

To capture the uncertainty in users' smartphone usage behavior, we do not fix the value of $\pi_s$ but instead perform random sampling on $\pi_s$ (selecting a set of $\pi_s$ that satisfy $\sum_s \pi_s = 1$). This approach yields a different TTE for each sample, ultimately producing a set of TTE. The simulation results are shown in Figure 10.

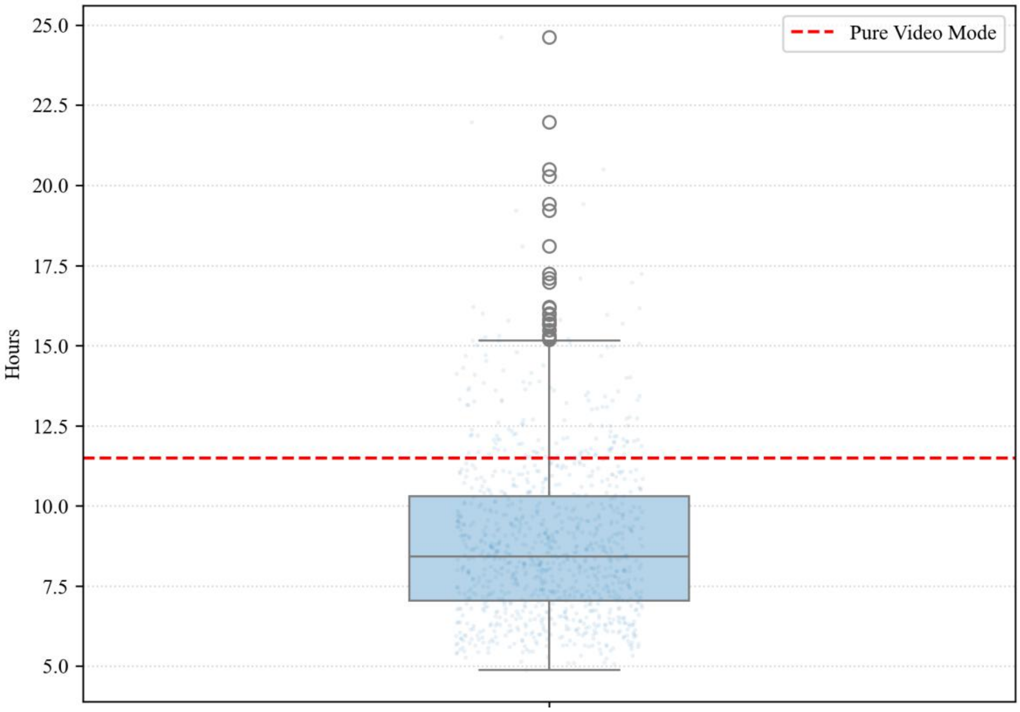


**Figure 10: M1: TTE Distribution under Mixed Mode Fluctuations (SOC=60%)**

The figure shows the M1 chip's predicted TTE under steady-state load at 11.5 hours (red dashed line). And the box plot illustrates the statistical distribution of TTE across 1,000 random fluctuation simulations. The median is approximately 8.8 hours, significantly lower than pure video mode, which indicates that during random usage, “high-power tasks” impact TTE far more than “low-power tasks”, signifying that switching between usage patterns greatly affects battery TTE. Simultaneously, the wide data distribution range (extending from approximately 5 hours to over 15 hours) authentically reflects the randomness inherent in user habits. Another interesting observation is that low-endurance data points cluster densely around 5 hours, while high-endurance data exhibits numerous outlier long tails. This reflects a fundamental physical reality of battery consumption: becoming power-hungry is easy (simply enable heavy loads), but achieving power efficiency faces physical limits (constrained by baseline standby power consumption).

# 8 Model Evaluation and Further Discussion

## 8.1 Strengths

- Strengths 1: The model accounts for multiple factors affecting battery performance, including usage modes, temperature, battery capacity, and aging level, enabling a comprehensive assessment of battery behavior.

- Strengths 2: We employ continuous-time differential equation modeling to accurately describe the trend of battery charge over time, providing a scientific basis for battery endurance prediction.
- Strengths 3: The model features a clear structure with well-defined physical parameters, facilitating understanding and application while maintaining strong scalability.
- Strengths 4: The model accounts for the nonlinear impact of temperature on battery performance, more realistically reflecting battery behavior under varying environmental conditions.
- Strengths 5: The model supports power consumption analysis across multiple usage modes, providing users with targeted battery optimization recommendations.

### 8.2 Weaknesses

- Weakness 1: The model assumes constant battery internal resistance and voltage, whereas actual battery internal resistance and voltage vary with SOC, temperature, and aging, potentially leading to prediction errors.
- Weakness 2: The model does not account for the recovery effect, i.e., the slight rebound in battery capacity during low-power states.
- Weakness 3: The model does not account for potential mutual power interactions among components.

### 8.3 Further Discussion

1. Consider variations in battery internal resistance
2. Consider voltage fluctuations during battery operation
3. Consider the battery recovery effect
4. Distinguish power consumption changes across different usage stages

## 9 Conclusion

The multi-factor coupled SOC model constructed in this study reveals that smartphone battery depletion is the result of the dynamic interaction of battery aging, ambient temperature, and real-time power consumption. This provides scientific support for addressing battery life anxiety.

The findings of the study indicate that battery degradation follows an "initially rapid, then gradual" decline pattern, with optimal performance observed around 25°C. Network signal quality and screen brightness have been identified as the predominant factors influencing power consumption, thereby challenging the prevailing notion that background applications are the primary contributors to energy expenditure.

This finding provides a critical insight: addressing challenges related to battery endurance necessitates concerted efforts across hardware, system, and user dimensions. Hardware should be designed to enhance thermal design in order to mitigate the effects of high-temperature aging. Systems must employ intelligent scheduling in order to dynamically adapt network modes and operational power consumption. Users can extend battery life through simple actions such as optimizing display settings and avoiding deep discharges.

Presently, as novel technologies such as foldable screens and artificial intelligence (AI) features become pervasive, device power demands will continue to escalate, while advancements in battery energy density will encounter constraints. Consequently, customized battery management solutions grounded in this paradigm may emerge as pivotal to achieving a balance between performance and endurance, thereby charting a novel course for the sustainable advancement of mobile devices.

# References

[1] Kemp, Simon. 2025. Digital 2026: Global Overview Report.DataReportal. https://datareportal.com/reports/digital-2026-global-overview-report.
[2] Kang, Minsoo. “Smartphone Replacement Cycle Forecast, 2016-2029.” Report. Counterpoint Research, November 19, 2025. https://www.counterpointresearch.com/insight/smartphone-replacement-cycle-forecast-2016-2029
[3] Ramadass, P., Bala Haran, Parthasarathy M. Gomadam, Ralph E. White, and Branko N. Popov. “Development of First Principles Capacity Fade Model for Li-Ion Cells.” Journal of The Electrochemical Society151, no. 2 (2004)
[4] Khan, S. M. Z., Alam, M. M., Le Moullec, Y., Kuusik, A., Parand, S., & Verikoukis, C. (2021). An empirical modeling for the baseline energy consumption of an NB-IoT radio transceiver. IEEE Internet of Things Journal, 8(19), 14756–14772.
[5] Apple Inc. (2023). iPhone 15 Pro Max Battery and Performance Technical Documentation. Available at: https://support.apple.com/kb/SP901
[6] Golden Reviewer. (2023). Genshin Impact Smartphone Power Consumption and Thermal Throttling Test. [Online Benchmarks].
[7] NotbookCheck. (2023). Hearthstone: Heroes of Warcraft - System Requirements and Battery Performance on Mobile Devices.
[8] GSMArena. (2023). Smartphone Battery Life Endurance Rating Methodology. Available at: https://www.gsmarena.com/battery-test.php3
[9] Zhag, L., et al. (2018). Energy Consumption Modeling for Mobile Displays: A Comprehensive Study. IEEE Transactions on Mobile Computing. DOI: 10.1109/TMC.2017.2788715.

# Report on Use of AI Tools

1. Deepl Translator (Aug, 2017 version, 5.1.3)
   Uploaded selected paragraphs writen in Mandarin to be translated into English.

2. GitHub CoPilot (Jan 16, 2024 version)
   Auto-completions for code used in preparing our models.

3. Deepl Write (Aug, 2017 version, 5.1.3)

Query:As shown in the figure, GPS power consumption (K5) and ambient temperature are the two factors that have the greatest impact on endurance prediction. Scenarios involving continuous GPS usage, such as navigation, have a significant negative impact on battery endurance. Near the analysed reference point, an increase in ambient temperature (within a reasonable range) helps to extend TTE. The elasticity of the ageing coefficient ( β ) is close to zero, indicating that, within the analytical framework of short-term endurance prediction, ageing has a much smaller impact on instantaneous TTE than real-time power consumption and ambient temperature.

Output:As shown in the figure, GPS power consumption (k5) and ambient temperature are the most critical factors affecting endurance prediction. Scenarios involving continuous GPS usage, such as navigation, have an extremely significant negative impact on battery endurance. Near the analyzed reference point, an increase in ambient temperature (within a reasonable range) helps extend TTE. The elasticity of the aging coefficient (β) is close to zero, indicating that within the analytical framework of short-term endurance prediction, aging has a far smaller impact on instantaneous TTE than real-time power consumption and ambient temperature.

4. Deepl Write (Aug, 2017 version, 5.1.3)

Query:The figure shows the predicted TTE of the M1 chip under a steady-state load of 11.5 hours (red dashed line). The box plot illustrates the statistical distribution of TTE across 1,000 random fluctuation simulations. The median is approximately 8.8 hours, which is significantly lower than in pure video mode. This indicates that, during random usage, 'high-power tasks' impact TTE far more than 'low-power tasks', showing that switching between usage patterns has a significant effect on battery TTE. The wide range of data (extending from approximately 5 to over 15 hours) accurately reflects the randomness of user habits. Another interesting observation is that low-endurance data points cluster densely around 5 hours, while high-endurance data exhibits numerous long tails of outliers. This reflects a fundamental physical reality of battery consumption: it is easy to become power-hungry (simply enable heavy loads), but achieving power efficiency faces physical limits (constrained by baseline standby power consumption).

Output:The figure shows the M1 chip's predicted TTE under steady-state load at 11.5 hours (red dashed line). And the box plot illustrates the statistical distribution of TTE across 1,000 random fluctuation simulations. The median is approximately 8.8 hours, significantly lower than pure video mode, which indicates that during random usage, “high-power tasks” impact TTE far more than “low-power tasks”, signifying that switching between usage patterns greatly affects battery TTE. Simultaneously, the wide data distribution range (extending from approximately 5 hours to over 15 hours) authentically reflects the randomness inherent in user habits. Another interesting observation is that low-endurance data points cluster densely around 5 hours, while high-endurance data exhibits numerous outlier long tails. This reflects a fundamental physical reality of battery consumption: becoming power-hungry is easy (simply enable heavy loads), but achieving power efficiency faces physical limits (constrained by baseline standby power consumption).